

calc.xlsx — sheet Inputs

Parameter	Value	Unit	Source
Upper voltage cut-off	4.7	V	LNMO.json
Lower voltage cut-off	2.5	V	LNMO.json
Positive stoichiometry limits (reaction window)	[0.9, 0.3]	-	LNMO.json
Nominal cell capacity (reference)	4.5	Ah	LNMO.json
Positive max concentration c_s,max	43000	mol/m3	LNMO.json
Negative max concentration c_s,max	33133	mol/m3	Chen2020.py:271
Positive density	4400	kg/m3	LNMO.json
Negative density	1657	kg/m3	Chen2020.py:283
Separator density	397	kg/m3	Chen2020.py:308
Positive AMVF	0.665	-	LNMO.json
Negative AMVF	0.75	-	Chen2020.py:275
Positive porosity	0.335	-	Chen2020.py:292
Negative porosity	0.25	-	Chen2020.py:274
Separator porosity	0.47	-	Chen2020.py:306
Electrode height	0.065	m	Chen2020.py:253
Electrode width	1.58	m	Chen2020.py:254
Positive electrode thickness	6e-05	m	params/V27_combo.json (60 um)
Negative electrode thickness	0.000109	m	params/V27_combo.json (109 um)
Separator thickness	1.2e-05	m	Chen2020.py:250
Positive current collector thickness	1.6e-05	m	Chen2020.py:252
Negative current collector thickness	1.2e-05	m	Chen2020.py:248
Initial concentration, negative electrode	29866	mol/m3	Chen2020.py:327
Initial concentration, positive electrode	17038	mol/m3	Chen2020.py:328
Initial concentration, electrolyte	1000	mol/m3	Chen2020.py:312
EC initial concentration in electrolyte	2000	mol/m3	params/V27_combo.json (base 4541, Chen2020.py:241)
Electrolyte conductivity	2	S/m	params/V27_combo.json (base: Nyman2008 fn)
Cation transference number	0.5	-	params/V27_combo.json
SEI kinetic rate constant	5e-14	m/s	params/V27_combo.json
Initial SEI thickness	5e-09	m	Chen2020.py:240
Total heat transfer coefficient	250	W/m2/K	params/V27_combo.json

calc.xlsx — sheet CapacityEnergy

Item	Value	Unit	Formula / basis	Source
1C discharge capacity	5.05694322191333	Ah	integral of 1C current over discharge	r10_V27_combo_1c.json capacity_ah
Discharge energy	21.11246886967024	Wh	integral V*dq	r10_V27_combo_energy.json energy_wh
Midpoint voltage (plateau metric)	4.155124697888494	V	voltage at half discharge energy (calc-energy contract)	r10_V27_combo_energy.json midpoint_voltage_v
DC resistance	0.004986278706756843	ohm	calc-energy contract	r10_V27_combo_energy.json dcr_ohm
Power density	18544.4248662352	W/kg	calc-energy contract	r10_V27_combo_energy.json power_density_w_kg
4C current (nominal basis)	18	A	4 x 4.5 Ah	nominal capacity x C-rate

calc.xlsx — sheet EnergyDensity

Item	Value	Unit	Formula	Source
Electrode sheet area	0.1027	m2	height x width = 0.065 x 1.58	Chen2020.py:253-254
Cell stack thickness	0.000209	m	16+60+12+109+12 um	layer thicknesses
Stack volume	2.14643e-05	m3	thickness x area	contract formula (bda energy.py)
Volumetric energy density	983.6085439390167	Wh/L	energy_wh / volume_m3 / 1000	r10_V27_combo_energy.json energy_density_wh_l
Positive layer mass	0.018030012	kg	L x AMVF x rho x A = 60e-6 x 0.665 x 4400 x A	calculated
Negative layer mass	0.013911716325	kg	L x AMVF x rho x A = 109e-6 x 0.75 x 1657 x A	calculated
Separator mass	0.0002593092840000001	kg	L x (1-por) x rho x A = 12e-6 x 0.53 x 397 x A	calculated
Al collector mass	0.00443664	kg	L x rho x A	calculated
Cu collector mass	0.011042304	kg	L x rho x A	calculated
Dry cell mass (contract)	0.04767998160900001	kg	sum of 5 layers	r10_V27_combo_energy.json mass_kg

Item	Value	Unit	Formula	Source
Gravimetric energy density	442.7952393690747	Wh/kg	energy_wh / mass_kg	r10_V27_combo_energy.json energy_density_wh_kg
Electrolyte mass (out of contract)	0.0065304876	kg	pore volume x 1.2 g/mL default	calculated; annotated default

calc.xlsx — sheet NPBalance

Item	Value	Unit	Formula	Source
Positive areal capacity	45.98330114233333	Ah/m2	L_p x eps_p x c_p,max x F / 3600	calculated
Negative areal capacity	72.59510147125084	Ah/m2	L_n x eps_n x c_n,max x F / 3600	calculated
N/P ratio	1.578727487322959	-	areal_neg / areal_pos = (L_n eps_n c_n,max)/(L_p eps_p c_p,max)	first-principles layer-capacity ratio
Design intent	plating clearance	-	oversized anode + thin cathode collapse 4C CC phase (~16.2 ms); DFN min anode pot. +0.1050 V	mechanism, r10 eval note

calc.xlsx — sheet FastChargeAging

Item	Value	Unit	Formula / basis	Source
4C CC charge accepted	8.086210729793998e-05	Ah	CC phase ends at 4.7 V cut-off from initial state 4.1995 V	r10_V27_combo_4c.json capacity_ah
4C CC phase duration	0.016172421459588	s	Q4C / 18 A	derived
Min anode potential, 4C	0.104950203915436	V	min over charge trajectory	r10_V27_combo_4c.json anode_potential_v
Plating verdict	False	-	min anode potential >= 0 V	log-evaluate r10 (plated auto-derived)
T_max, 4C charge	318.9224559656013	K	max over charge trajectory	r10_V27_combo_4c.json T_max_K
T_max, 1C discharge	299.0339593147587	K	max over discharge trajectory	r10_V27_combo_1c.json T_max_K
SEI thickness, 100 cycles	313.6994640602236	nm	Yang2017 EC-reaction-limited growth	r10_V27_combo_aging.json sei_thickness_nm_end
SEI growth over 100 cycles	308.6994640602236	nm	end - initial (5 nm)	derived
Capacity cycle 1 / cycle 100	-0.618 / 2.732	Ah	aging trajectory	r10_V27_combo_aging.json capacity_ah_per_cycle
Trajectory note	SEI-model lithium-inventory climb artifact	-	disclosed; durable metric = SEI thickness	honesty annotation
Margin vs threshold: ED	33.60854393901673	Wh/L	983.61 - 950	r10 evaluate
Margin vs threshold: plateau	0.05512469788849472	V	4.15512 - 4.1	r10 evaluate
Margin vs threshold: T_max	4.227544034398704	K	323.15 - 318.92	r10 evaluate
Margin vs threshold: anode potential	0.104950203915436	V	+0.10495 - 0	r10 evaluate
Margin vs threshold: SEI	186.3005359397764	nm	500 - 313.70	r10 evaluate